MDTS4214_729_PSET6

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#*To demonstrate the application of subset selection methodsConsider the “Hitters” data. For loading the data you need to install ISLRlibrary. We wish to predict a baseball player’s salary on the basis of various predictors. So first we need to choose an optimal model.1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
library(ISLR)

## Warning: package 'ISLR' was built under R version 4.5.2

data(Hitters)

# Remove missing values
Hitters <- na.omit(Hitters)
```

*2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```
set.seed(1)

train_index <- sample(1:nrow(Hitters), 0.8*nrow(Hitters))

train <- Hitters[train_index, ]
test  <- Hitters[-train_index, ]
```

*3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC , AIC and Cp values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

```
library(leaps)

## Warning: package 'leaps' was built under R version 4.5.3
```

```r
best_model <- regsubsets(Salary ~ ., data = train, nvmax = 19)

summary_best <- summary(best_model)
# Adjusted R² (maximum)
which.max(summary_best$adjr2)

## [1] 11

# BIC (minimum)
which.min(summary_best$bic)

## [1] 5

# Cp (minimum)
which.min(summary_best$cp)

## [1] 6
```

*4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```r
predict_regsubsets <- function(object, newdata, id){
  form <- as.formula(object$call[[2]])
  mat <- model.matrix(form, newdata)
  coefi <- coef(object, id=id)
  vars <- names(coefi)
  pred <- mat[, vars] %*% coefi
  return(pred)
}

# Example: model with k variables
k <- which.min(summary_best$bic)

# Train MSE
train_pred <- predict_regsubsets(best_model, train, id=k)
train_mse <- mean((train$Salary - train_pred)^2)

# Test MSE
test_pred <- predict_regsubsets(best_model, test, id=k)
test_mse <- mean((test$Salary - test_pred)^2)

train_mse

## [1] 91671.96

test_mse
```

```
## [1] 182578.9
```

*5. Repeat 3-4 using forward stepwise selection method.6. Repeat 3-4 using backward stepwise selection method.1 7. Compare the results of the three methods in terms of choice of best model,train MSE amd test MSE.

```
forward_model <- regsubsets(Salary ~ ., data = train,
                            nvmax = 19, method = "forward")

summary_forward <- summary(forward_model)

backward_model <- regsubsets(Salary ~ ., data = train,
                             nvmax = 19, method = "backward")

summary_backward <- summary(backward_model)
summary_backward

## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train, nvmax = 19, method = "b
ackward")
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: backward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns
```

```
CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    "*"   " "    " "
" "
## 2  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    "*"   " "    " "
" "
## 3  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    "*"   " "    " "
" "
## 4  ( 1 )  " "   " "  " "   " "  " " " "   "*"   " "    "*"   " "    " "
" "
## 5  ( 1 )  " "   "*"  " "   " "  " " " "   "*"   " "    "*"   " "    " "
" "
## 6  ( 1 )  "*"   "*"  " "   " "  " " " "   "*"   " "    "*"   " "    " "
" "
## 7  ( 1 )  "*"   "*"  " "   " "  " " "*"   "*"   " "    "*"   " "    " "
" "
## 8  ( 1 )  "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
" "
## 9  ( 1 )  "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 14  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   " "    " "
"*"
## 15  ( 1 ) "*"   "*"  " "   " "  "*" "*"   "*"   "*"    "*"   " "    " "
"*"
## 16  ( 1 ) "*"   "*"  " "   "*"  "*" "*"   "*"   "*"    "*"   " "    " "
"*"
## 17  ( 1 ) "*"   "*"  " "   "*"  "*" "*"   "*"   "*"    "*"   " "    "*"
"*"
## 18  ( 1 ) "*"   "*"  " "   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"
"*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"
"*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       "*"     " "     " "    " "
## 3  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 4  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 9  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 10  ( 1 ) " "    " "     "*"       "*"     "*"     " "    " "
```

```
## 11  ( 1 ) " "    " "     "*"       "*"     "*"     "*"    " "
## 12  ( 1 ) " "    "*"     "*"       "*"     "*"     "*"    " "
## 13  ( 1 ) " "    "*"     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
```

Final Conclusion (Model Selection – Hitters Data) Different selection methods (Best Subset, Forward, Backward) may choose different models depending on criteria like Adjusted □2R2, BIC, and Cp.BIC usually selects a simpler model (fewer variables), while Adjusted □2R2 often prefers more complex models. When comparing performance: Training MSE is always lower for more complex models. But the best model is the one with the lowest Test MSE, not training MSE.